

Tank Hardware List Workbook — Designer & Developer Reference

Oct 9, 2026

Overview

The workbook turns a transformer job's joint list into three outputs: a purchase summary (FINAL SHEET), a shopfloor fitting list (PRODUCTION SHEET) and a standards check (VERIFICATION). Designers type into two input sheets; one REFRESH button rebuilds everything else.

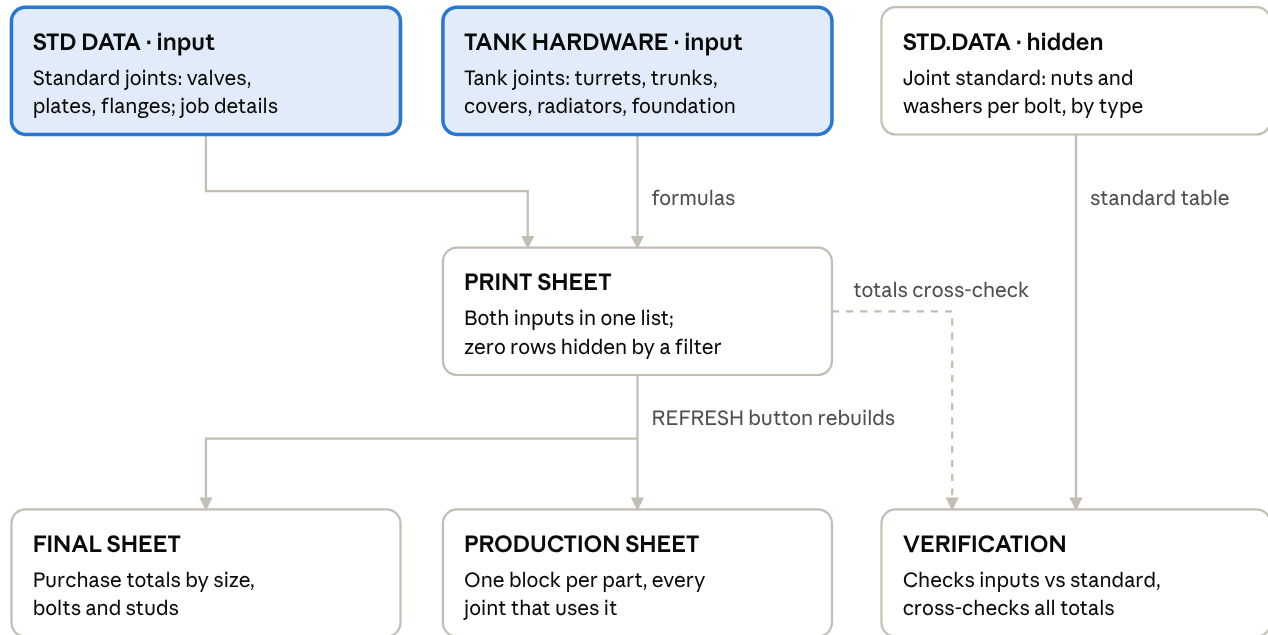
Reader	Read these sections
Designer (enters job data, prints, checks)	Sheet map, Designer workflow, How VERIFICATION works, Troubleshooting
Developer (maintains the macros)	All sections, especially REFRESH, PRODUCTION SHEET build, Developer reference, Maintenance rules

- **File in use:** Document/New Feauture/6367_HARDWARE LIST_PRODUCTION.xlsm (WO 6367). Built from the designers' 6367_HARDWARE LIST.xlsm by the installer script; the designers' file is never edited.
- **Format:** F/DSN/54, version 1.4. Macro-enabled (.xlsm) — saving as .xlsx deletes every macro.
- **WO 6367 after REFRESH:** 1,912 bolts, 118 studs, 2,542 nuts, 4,464 plain washers, 634 spring washers, 31 lock nuts; PRODUCTION SHEET holds 28 part blocks on 4–5 printed pages.

Sheet map

Designers type in two sheets; everything else is built from them

Shaded = input sheets. Arrows = where data flows.



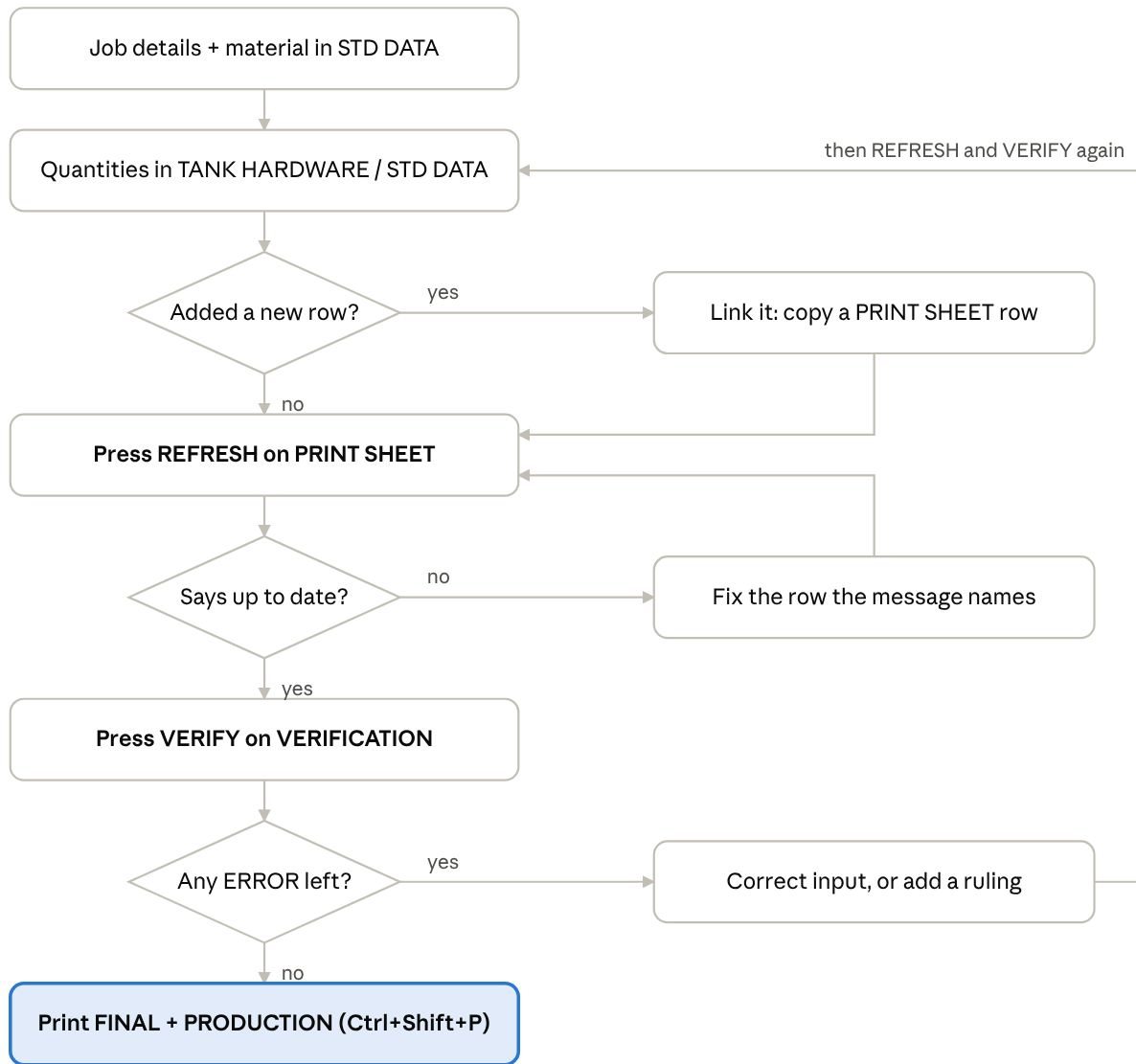
sheet map · 2 input sheets, 1 hidden standard, 4 built sheets

PRINT SHEET joins both input sheets by formulas; the **REFRESH** button then rebuilds **FINAL** and **PRODUCTION SHEET** from it. **VERIFICATION** reads the inputs directly against the hidden standard table and checks that all totals agree.

Designer workflow

Ready to print once REFRESH is clean and VERIFY shows no errors

Designer workflow for one job



designer workflow · 5 steps, 3 checks

PRINT SHEET follows the inputs on its own; FINAL and PRODUCTION SHEET change only when REFRESH runs (or when they are printed with Ctrl+Shift+P/Q, which rebuilds first). VERIFY is a separate button and is never run by REFRESH.

Sheet-by-sheet reference

Only TANK HARDWARE and STD DATA take typing; every other visible sheet is formulas or macro output.

Sheet	Who writes it	Holds	Key columns / rows
STD DATA	Designer	Job details (WO C1, rating C2, customer C3, material ≤M12 C4, >M12 C5, revision C6) and the standard joints: valves, plates, flange joints	Same columns as TANK HARDWARE; joint rows 15–95
TANK HARDWARE	Designer	Tank-specific joints (turrets, trunks, covers, radiators, foundation bolts). C1–C6 copy STD DATA	B location · C qty/input · D joint type · F size · G–I plate/gasket/plate · K required length · L chosen length · M part name · N bolt/stud · O nut · P lock nut · Q plain washer · R spring washer; rows 15–250
PRINT SHEET	Formulas	One combined list of both input sheets, in print order; a filter on column B hides zero rows	D location · F part name · G bolt/stud · H nut · I P.washer · J S.washer · K L.nut · L STD.REF; rows 9–326. REFRESH button at N2
FINAL SHEET	Macro RebuildFinalSheet	Purchase totals: nuts/washers by size + material, then bolts, then studs	Blocks at rows 11–34, 37–96, 99–125

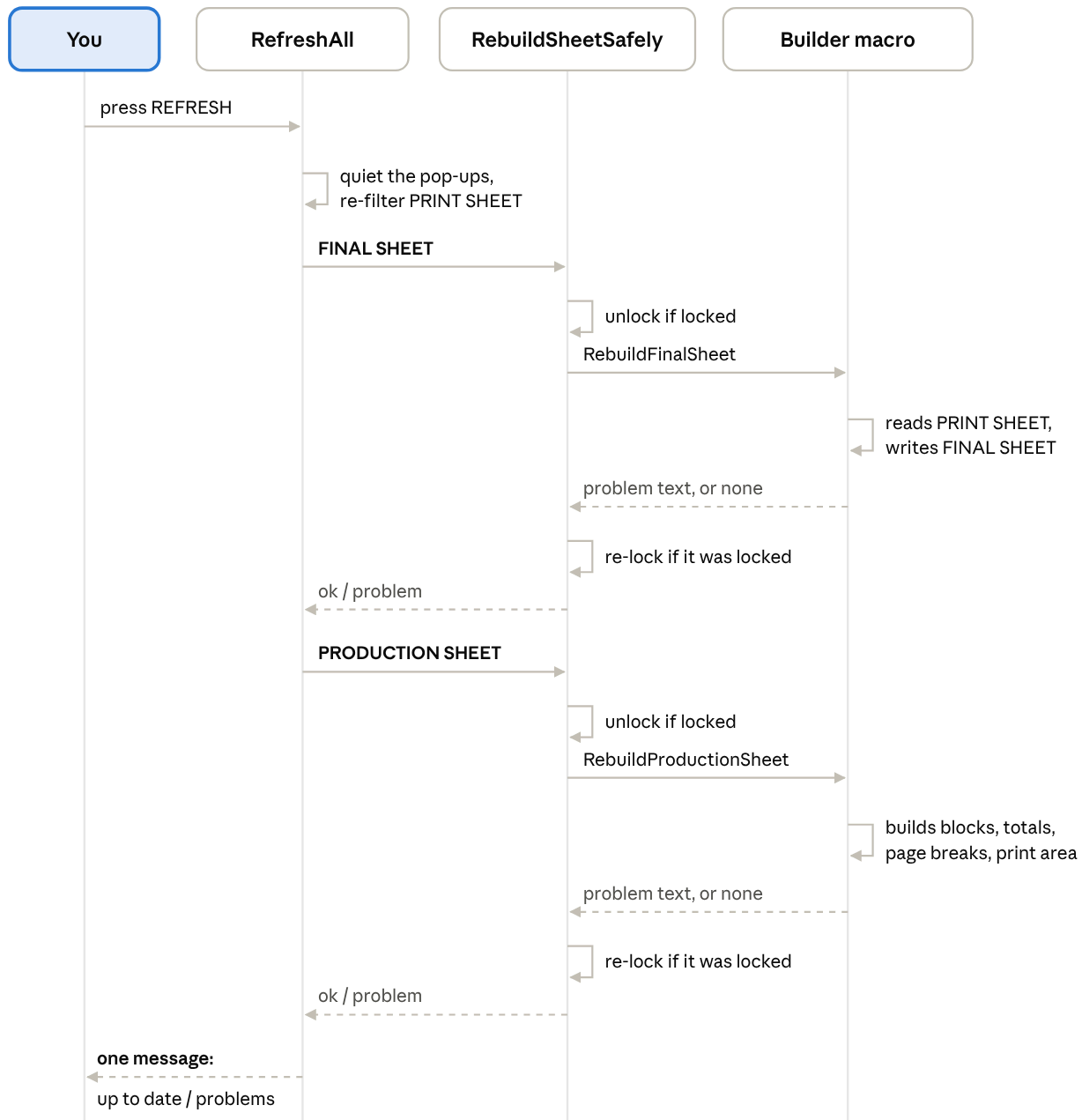
Sheet	Who writes it	Holds	Key columns / rows
PRODUCTION SHEET	Macro <code>RebuildProductionSheet</code>	Shopfloor list: one yellow block per part with every joint location that uses it and a TOTAL	Header rows 2–6 fixed; blocks from row 7 down
VERIFICATION	Macro <code>RunVerification</code>	Deviations from the joint standard, plus total cross-checks	RULINGS rows 14–23 · summary 27–30 · findings 34–183. VERIFY button
STD.DATA (hidden)	Designer, rarely	The joint standard table: per bolt, how many nuts and washers each joint type needs	Header <code>REF-STD</code> in column K (row 220); entries SD-267 ... SD-274 below it
PRINT QTY, NON.STD.DATA (hidden)	—	Legacy sheets, not used by the macros	—

Note the two similar names: **STD DATA** (space, visible) is an input sheet; **STD.DATA** (dot, hidden) holds the standard table.

What REFRESH does inside

One REFRESH press rebuilds FINAL, then PRODUCTION, then shows one message

Solid = call, dashed = answer back, loop = work done inside that macro



REFRESH call sequence · 4 macros, 2 rebuilds

A sheet locked with a password is never unlocked (Excel is not allowed to prompt); it is reported instead. The print shortcuts take the same path for the one sheet being printed:

`PRINTOUTPUT` / `PRINTPROTECTEDOUTPUT` call `RebuildSheetSafely`, and print only if it answers ok — so stale paper never reaches the shopfloor.

How PRODUCTION SHEET is built

Nothing on the sheet changes until every part name has been checked

RebuildProductionSheet, step by step



RebuildProductionSheet · read-only phase, then write phase

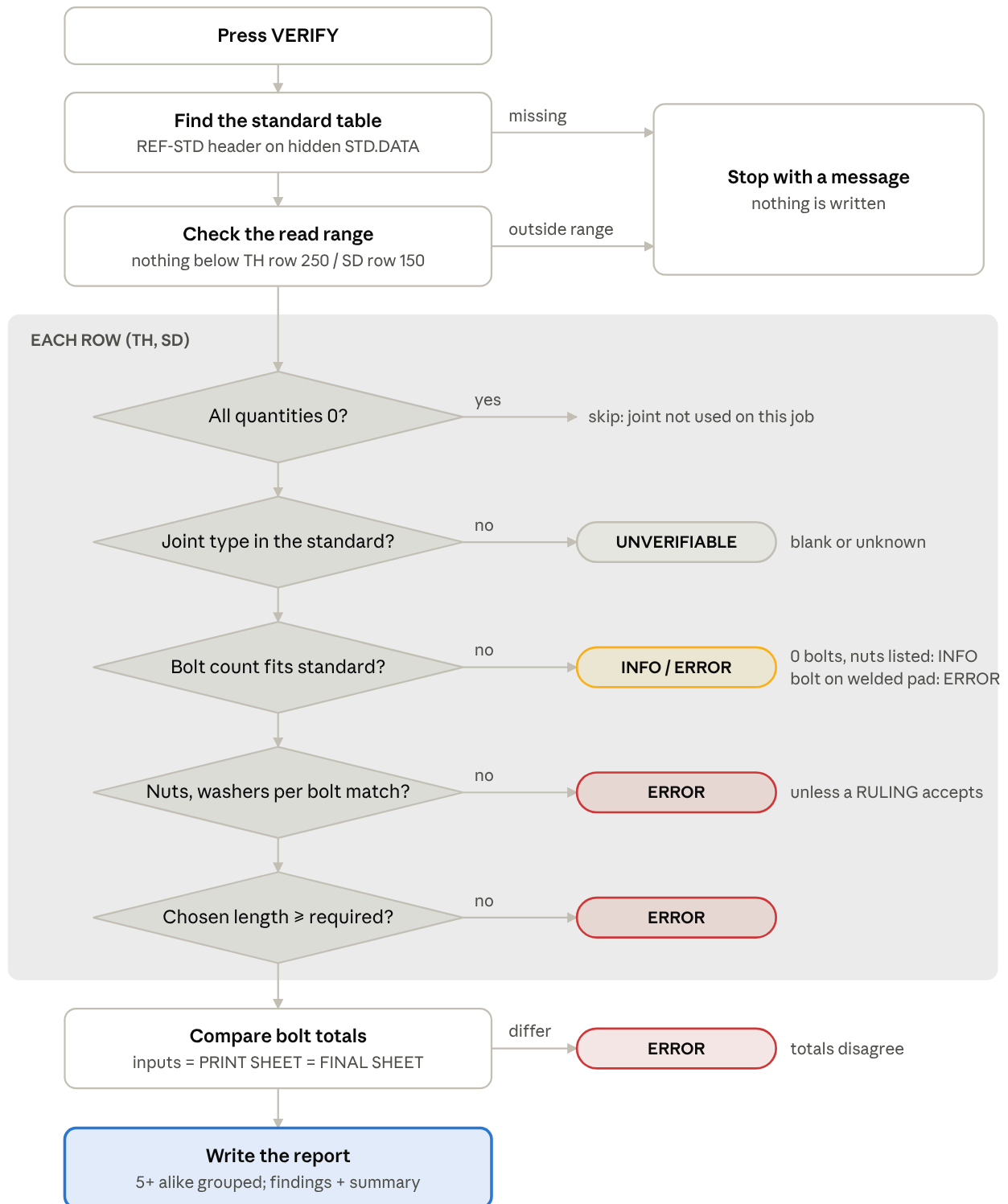
Keep blocks whole. Excel has no keep-together setting, so the macro asks Excel where each page would end (`HPageBreaks`). If a page would end inside a block, or within `SAFETY_ROWS` (2) rows of its TOTAL, it adds a page break above that block's title and asks again. A block taller than a whole page starts on a fresh page and is split only where it must be.

Grouping key. The part name is rebuilt from its pieces — size, `-STUD` if a stud, length, material — the same way FINAL SHEET reads it, so both sheets always split parts identically.

How VERIFICATION works

Every used row is checked against the standard, then the totals

RunVerification. Chips = severity written to the FINDINGS list.



RunVerification · 2 guards, 5 row checks, 1 total check

A "yes" at each diamond moves down to the next check; the row checks run for every used row of both input sheets. One more INFO check runs on every row: if a joint's name mentions a size ("FOUNDATION BOLT - M24x300") that differs from its part (M24x320), it is noted — foundation bolts are deliberately 20 mm longer for grouting.

Severity	Means	Count for WO 6367
ERROR	Contradicts the standard; fix it or accept it with a ruling	8
INFO	Needs a human eye, not necessarily wrong	9
UNVERIFIABLE	No standard to compare with (blank or unknown joint type)	8

Rulings (VERIFICATION rows 14–23) accept a deviation: B = joint type (SD-268), C = NUT , P.WASHER or S.WASHER , D = the accepted ratio; E–G are a record only. A ruling applies to every row of that joint type on both input sheets, and cannot silence BOLT, LENGTH, TOTAL or UNVERIFIABLE findings.

Developer reference

Four VBA modules do all the work; the working sources, installer and tests live in `.claude/route/` beside the workbook folder.

Module	Public macros	Called by
ProductionSheetBuilder	RefreshAll , RebuildSheetSafely , RebuildProductionSheet , SetProductionSilent , ProductionLastProblem	REFRESH button (PRINT SHEET), print macros
FinalSheetBuilder	RebuildFinalSheet , RefreshFinalSheetVisibility (old name), SetSilentMode , LastProblem	RebuildSheetSafely
VerificationBuilder	RunVerification , SetVerifySilent , VerifyLastProblem	VERIFY button

Module	Public macros	Called by
Module1	PRINTOUTPUT (Ctrl+Shift+P), PRINTPROTECTEDOUTPUT (Ctrl+Shift+Q)	Keyboard, PRINT OUTPUT PAGE button

Constants worth knowing

Constant	Module	Value	Meaning
PD_FIRST_ROW	ProductionSheetBuilder	7	First row PRODUCTION SHEET blocks may use
PD_TITLE_ROWS	ProductionSheetBuilder	\$2:\$6	Header repeated on every printed page
SAFETY_ROWS	ProductionSheetBuilder (KeepBlocksTogether)	2	Spare rows kept at a page bottom; raise by 1 if a TOTAL row spills
FS_B1_*, FS_B2_*, FS_B3_*	FinalSheetBuilder	11-34, 37- 96, 99- 125	FINAL SHEET block rows
TH_FIRST / TH_LAST / TH_GUARD	VerificationBuilder	15 / 250 / 320	TANK HARDWARE rows read, and guard zone
SDI_FIRST / SDI_LAST / SDI_GUARD	VerificationBuilder	15 / 150 / 220	STD DATA rows read, and guard zone
SD_REF_HEADER	VerificationBuilder	REF-STD	Header that locates the standard table
GROUP_MIN	VerificationBuilder	5	Findings shared by this many rows are grouped

Changing a macro safely

1. Edit the .bas file in .claude/route/, never line-patch inside Excel by code (one DeleteLines call once wiped a whole module).

2. Rebuild the output: `python .claude/route/install_production.py "<copy of designers' file>.xlsm"`. It imports all four modules whole (keeps the Ctrl+Shift+P/Q shortcut lines), moves the REFRESH button, links PRINT SHEET rows for TANK HARDWARE 59/60 if missing, sets the page footers, unhides VERIFICATION, runs VERIFY and REFRESH.
3. Run the tests on scratch copies (each opens its own hidden Excel and never saves the original):

Script	Proves
<code>verify_production.py</code>	PRODUCTION SHEET equals a recomputation from PRINT SHEET; totals equal FINAL SHEET
<code>flow_tests.py</code>	Designer flow: qty change, joint on/off, new size, hidden-row print path, HDG→SS spec
<code>edge_tests.py</code>	Idempotent, locked/password sheets, bad part names, shrinking list, settings restored
<code>verification_tests.py</code>	STD DATA rows checked, rulings, totals, table drift, error values, range guard
<code>check_pagebreaks.py</code>	No block split across printed pages (set the copy to Letter paper first — see Maintenance rules)

Maintenance rules and known gaps

Most past bugs came from rows that moved or hid; these rules keep them from coming back.

Rules

- **New input row → new PRINT SHEET row.** A row added to TANK HARDWARE or STD DATA reaches FINAL and PRODUCTION SHEET only through a PRINT SHEET row that links to it. Copy the neighbouring PRINT SHEET row and insert it below, so the formulas shift to the new row. VERIFICATION reports a forgotten link as "totals disagree". (TANK HARDWARE rows 59 and 60, added in revision R2, were missed this way: 16 bolts.)
- **Never find the end of PRINT SHEET with `End(xlUp)` or `Find`.** Its filter hides zero rows and both skip filtered rows, so a joint switched on at the bottom (the foundation bolts) vanished from both summaries. Use `UsedRange` as all three builders now do.

- **Never read a table by fixed row numbers.** The standard table moved from rows 222–227 to 221–228 when SD-267 and SD-274 were added; VERIFICATION now finds it by its REF-STD header.
- **Keep material in the part key.** M12x60-SS and M12x60-HDG are different parts on every sheet.
- **FINAL SHEET keeps merged cells**, so it is written by a macro, not by spilling formulas (Excel refuses #SPILL! into merged cells).

Known gaps

- **Rulings are joint-wide.** A VERIFICATION ruling (e.g. SD-268 / NUT = 2) applies to every row of that joint type on both input sheets. A row-scoped version exists in .claude/route/VerificationBuilder-2026-08-25.bas but needs a different RULINGS layout.
- **Blank joint types:** STD DATA rows 40–41 and 44–49 (inspection covers) have no joint type, so VERIFICATION cannot check them.
- **PDF on the development PC:** Excel's Save as PDF there always writes Letter pages, whatever the sheet says. Page-break tests set the copy to Letter so plan and PDF match. On the shop printer (A4) Excel's own page plan applies.
- **REFRESH does not run VERIFY.** Press VERIFY separately after input changes.

Troubleshooting

Every macro that stops says "Nothing has been changed" when it left the sheet as it was; fix what it names and run it again.

You see	Meaning	Fix
"FINAL SHEET and PRODUCTION SHEET are up to date."	REFRESH worked	Nothing
"could not read N part name(s) on PRINT SHEET"	A part name in PRINT SHEET column F is not like M12x50-HDG / M12-STUDx75-HDG , or shows #REF!	Correct the input row it names on TANK HARDWARE or STD DATA

You see	Meaning	Fix
"... is protected with a password"	The sheet is locked with a password, so the macro will not unlock it	Review › Unprotect Sheet, then REFRESH
PRODUCTION SHEET shows old numbers	Inputs changed, REFRESH not pressed	Press REFRESH (printing with Ctrl+Shift+P also rebuilds first)
Nothing printed after Ctrl+Shift+P	The rebuild before printing hit a problem and showed it	Fix it, print again
A TOTAL row prints on the next page	The printer fits a row less than Excel planned	Raise SAFETY_ROWS from 2 to 3
VERIFICATION: "totals disagree: A vs B"	Input sheets vs PRINT SHEET: a row is not linked. PRINT SHEET vs FINAL SHEET: REFRESH not pressed	Link the row (Maintenance rules) or press REFRESH
VERIFICATION: "joint type not in STD.DATA"	Column D blank, or a code the standard table lacks	Fill the joint type, or add the code under REF-STD on STD.DATA
VERIFICATION: "Could not find the STD.DATA standard table"	The REF-STD header in STD.DATA column K was removed or renamed	Restore the header
Buttons do nothing	Macros disabled, or saved as .xlsx	Enable Content; keep the .xlsm